# SPC - Most Common IATF Audit NC Findings

What auditors find on SPC charts and how to prevent every finding

QUALITY HEAD AGENTS | AIAG SPC 2nd Edition | IATF 16949:2016

## Major NC Findings - SPC

- NC: Control limits set equal to USL/LSL - most common Major NC in SPC
- NC: Control chart not updated at workstation (last entry 3 shifts ago)
- NC: Cpk < 1.33 for CC characteristic with no corrective action plan
- NC: Ppk not reported in PPAP Element 9 for CC characteristics
- NC: No reaction plan followed after out-of-control signal - production continued
- NC: SPC not applied to any CC characteristics in the Control Plan

## Minor NC Findings - SPC

- NC: Subgroup size inconsistent - some subgroups have 4 parts, some have 5
- NC: Control limits not recalculated after 4M change
- NC: Nelson rules not defined - operators do not know which rules to apply
- NC: No out-of-control event log - cannot demonstrate reactions to signals
- NC: SPC data not used in Management Review - no trend analysis presented
- NC: Cpk ongoing not tracked - only historical PPAP Ppk on file
- NC: Gauge GRR not completed for gauge used in SPC data collection

## Prevention Checklist

- Calculate control limits ALWAYS from process data - never from specs.
- Display control limits and spec limits separately on the chart.
- Define Nelson rules in the SPC plan - train all operators.
- Maintain an out-of-control log: date, rule, action, restart verification.
- Review Cpk monthly - include trend in Management Review.
- After every 4M change: recalculate control limits, run new capability study.
- Confirm GRR < 30% for every gauge used in SPC before starting chart.
- Include SPC scope in Control Plan: characteristic, chart type, frequency, gauge, who.